

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XVI

JEE (Main)-2022

TEST DATE: 07-07-2022

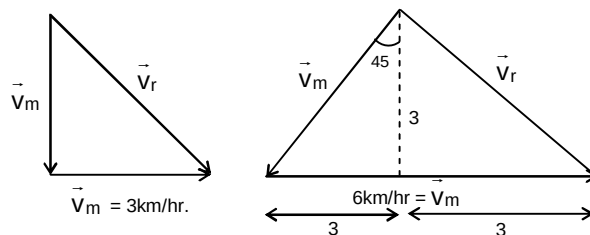
ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. A
Sol. $\vec{v}_r$ should be same in both the situations.
Thus, $|\vec{v}_r| = 3\sqrt{2}$ km/hr.



2. C
Sol. Power $P = F \cdot v = \frac{K}{v} \cdot v = K = \text{constant}$
 $\therefore W = P \cdot t = Kt$

3. B
Sol. $n_1 = k(l_1 - l)$
where l is its natural length
 $n_2 = k(l_2 - l)$
 $\frac{n_1}{n_2} = \frac{(l_1 - l)}{(l_2 - l)}$
On solving, $l = \frac{n_1 l_2 - n_2 l_1}{(n_1 - n_2)}$

$$\begin{aligned}
 \text{Now,} \quad n_1 &= k[(l_1 - l)] \\
 &= k \left[l_1 - \frac{(n_1 l_2 - n_2 l_1)}{(n_1 - n_2)} \right] \\
 k &= \frac{(n_1 - n_2)}{(l_1 - l_2)}
 \end{aligned}$$

4. B

 Sol. If initial velocities are: $u_1 = \sqrt{2gh}(\downarrow)$, $u_2 = \sqrt{2gh}(\uparrow)$

 Then final velocities: $v_1 = 3\sqrt{2gh}$, $v_2 = \sqrt{2gh}$

 By using conservation of momentum and equation of e & $m \ll M$.

5. C

$$\text{Sol. } m \frac{dv}{dt} = -6\pi\eta r v$$

$$(1) \frac{dv}{dt} = 6\pi \left(\frac{1}{18\pi} \right) (1) v$$

$$\frac{dv}{dt} = -\frac{v}{3}$$

$$\frac{dv}{v} = -\frac{dt}{3}$$

$$[\ln v]_2^{0.5} = \left[-\frac{t}{3} \right]_{t_1}^{t_2}$$

$$\ln 4 = \frac{\Delta t}{3}$$

$$\Delta t = 3 \ln 4$$

6. C

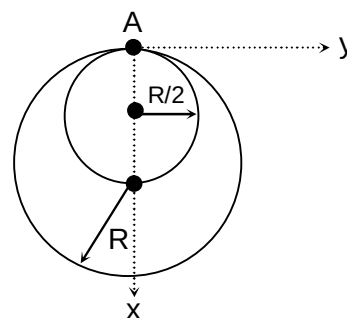
$$\text{Sol. } x_{cm} = \frac{m \left(\frac{R}{2} \right) + 2mR}{m + 2m} = \frac{5}{6} R$$

$$I_A = I_1 + I_2, \quad I_1 = m \left(\frac{R}{2} \right)^2 + m \left(\frac{R}{2} \right)^2 = \frac{mR^2}{2}$$

$$I_2 = 2mR^2 + 2mR^2 = 4mR^2 \Rightarrow I_A = \frac{9}{2} mR^2$$

$$\text{for compound pendulum } T = 2\pi \sqrt{\frac{I}{Mgd}}$$

$$\Rightarrow \text{ here } M = 3m, \quad d = \frac{5}{6} R \Rightarrow T = 2\pi \sqrt{\frac{9R}{5g}}$$



7. B

$$\text{Sol. } 0.6 = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{20}{Q_1}$$

$$\Rightarrow Q_1 = 50, W = Q_1 - Q_2 = 30 \text{ J}$$

8. C

Sol. On the horizontal, the path difference is decreasing so at the maximum distance from P where minima occurs, path difference is $\frac{\lambda}{2}$.

$$\text{Hence } \sqrt{9\lambda^2 + x^2} - x = \frac{\lambda}{2} \Rightarrow x = 8.75 \lambda$$

9. D

$$\text{Sol. } t = 20 \text{ min} = \frac{1}{3} \text{ hr}$$

$$1 \text{ km} = \sqrt{(6)^2 - V^2} \times \left(\frac{1}{3} \text{ hr}\right)$$

$$V = 3\sqrt{3} \text{ km/hr.}$$

10. A

$$\text{Sol. } 12.75 = \left| E_0 - \frac{E_0}{n^2} \right| = \left| -13.6 - \frac{(-13.6)}{n_1^2} \right|$$

$$\Rightarrow n = 4$$

$$\text{no. of lines} = \frac{n(n-1)}{2} = 6$$

11. B

$$\text{Sol. } qvB = \frac{mv^2}{R}$$

$$mv = qRB = 1.6 \times 10^{-21} \text{ kg m/sec}$$

12. D

$$\text{Sol. } E = \frac{\rho}{4\epsilon_0} r$$

$$V_{AB} = E \times \frac{r}{2}$$

13. B

$$\text{Sol. } \frac{E}{4} = 3$$

$$\Rightarrow E = 12$$

14. C

Sol. Let dq be the charge which has passed in a small interval of time dt, then dq = idt = (4 + 2t)dt. Hence total charge passed between interval t = 2 sec and t = 6 sec

$$q = \int_2^6 (4 + 2t)dt = 48 \text{ coulomb.}$$

15. D

Sol. Maximum stress on the wire will be at highest point (at the point of suspension)

$$\therefore \text{stress} = \frac{\text{weight}}{A} = \frac{A\rho g}{A} = \sigma$$

$$\therefore l = \frac{\sigma}{\rho g}$$

16. D

 Sol. I_L and I_C will be in opposite phases.

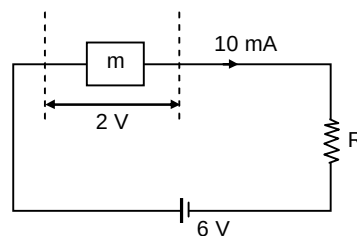
$$I_{\text{net}} = I_L - I_C \\ = 0.6 - 0.4 = 0.2 \text{ A}$$

17. A

 Sol. $\varepsilon = B\ell v$ develops across rod PQ.

By Lenz's law, plate M get positively charged.

18. D

 Sol. The potential drop across R is 4 V while current is 10 mA hence, $10 \times 10^{-3} \times R = 4$.
 $R = 400 \Omega$.


19. B

Sol. The electron density in doped semiconductor,

$$\eta_e = \frac{n_i^2}{n_h} = \frac{(3 \times 10^{16})^2}{4.5 \times 10^{22}} = 2 \times 10^{10} / \text{m}^3$$

20. B

 Sol. As accelerating voltage is increased, energy increases; the λ_{min} decreases and X-rays are getting harder (less wavelength) and penetrating power increases if filament current increases, more electrons are emitted.

SECTION – B

21. 00004.00

 Sol. $Td\theta = \lambda r d\theta \omega^2 r$

$$T = \lambda r^2 \omega^2$$

$$\text{Stress} = \frac{T}{A} = \frac{\lambda r^2 \omega^2}{A} \\ = 4$$

22. 00001.80

 Sol. $(T + T \cos 60) v_w - T v_b = 0$

23. 00041.40

Sol. $v_{\text{esc}} = \sqrt{\frac{2GM}{R}} ; v_{\text{orb}} = \sqrt{\frac{GM}{R}}$

24. 00004.00

Sol. $V = \frac{dx}{dt} = 2t, W = \frac{1}{2}mv^2 = \frac{1}{2} \times 0.5 \times (4)^2$

25. 00005.00

Sol. Friction force = $ma = 1 \times 5 = 5$.

26. 00037.25

Sol. $T + v\rho g = v\rho g$

$$T = v(\rho - \rho\omega)g$$

$$= \frac{4}{\rho}(\rho - \rho\omega)g = 37.25$$

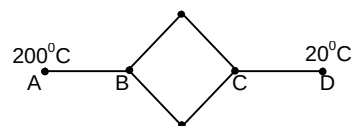
27. 00140.00

Sol. $Re_q = R + R + R = 3R$

$$H = \frac{200 - 20}{Re_q} = \frac{180}{3R} = \frac{60}{R}$$

$$200 - T_B = R \times H = 60$$

$$T_B = 140^\circ\text{C}$$



28. 00001.22

Range: 1.21 to 1.23

Sol. $\rho h^2 = \text{constant}$

29. 00005.67

Range: 5.62 to 5.70

Sol. distance = $\int_0^4 |t^2 - 5t + 6| dt$

30. 00000.33

Range: 0.32 to 0.34

Sol. $Kx = \frac{Mg}{3} ; x = \frac{Mg}{3k}$

Chemistry

PART – B

SECTION – A

31. D

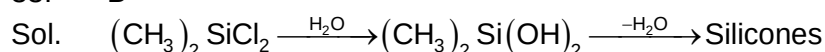
Sol. Sulphur forms two sigma bonds with two lone pairs.

32. B

Sol. At high pressure 'a' is neglected due to repulsion between gas molecules.

$$\therefore P(V - b) = RT \text{ or } PV = RT + Pb$$

33. B



34. D

Sol. $K_p = \frac{p_y \times p_z}{p_x} = \frac{3 \times 3}{3} = 3$

$$P_{\text{equilibrium}} = p_x + p_y + p_z = 3 + 3 + 3 = 9 \text{ atm}$$

35. B

Sol. $\frac{n_{\text{H}_2}}{n_{\text{O}_2}} = \sqrt{\frac{M_{\text{O}_2}}{M_{\text{H}_2}}}$

$$\text{or, } \frac{2/2}{10} \times \frac{10}{n_{\text{O}_2}} = \sqrt{\frac{32}{2}} = 4$$

$$\therefore \frac{1}{n_{\text{O}_2}} = 4 \Rightarrow n_{\text{O}_2} = \frac{1}{4} \Rightarrow W_{\text{O}_2} = \frac{1}{4} \times 32 = 8 \text{ g}$$

36. A



37. B

Sol. Zn^{2+} form colourless compounds in solution due to absence of d-d transition.

38. D

Sol. The products are due to coupling of different alkyl groups. If $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$ is RCl and $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_3$ is $\text{R}'\text{Cl}$, the products will be RR , $\text{R}'\text{R}'$ and RR' .



39. B

Sol. For first order reaction, the half-life time is given as: $t_{1/2} = \frac{0.693}{K}$

$\therefore$ The half-life is independent of the concentration of reactant.

40. B

Sol. $\frac{\Lambda_m}{\Lambda_e} = 3 \Rightarrow \Lambda_m = 3\Lambda_e$

n-factor of $\text{MgCl}_2 \cdot \text{KCl} \cdot 6 \text{H}_2\text{O} = 3$.

41. D

Sol. Due to weak Bi – H bond.

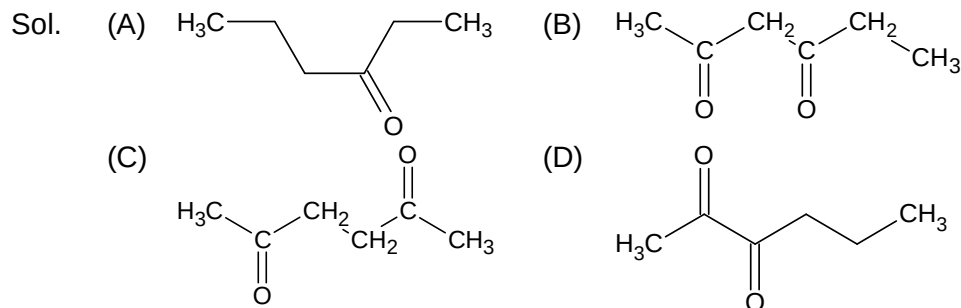
42. B

Sol. The diene is $\text{CH}_3\text{CH} = \text{CH} - \text{CH}_2 - \text{CH} = \text{CHCH}_3$

43. D

Sol. Xe undergoes sp^3 hybridization and forms three sigma and three pi-bonds with a lone pair on it.

44. B



45. B

Sol. It is a buffer of weak acid and a salt of strong base.

46. C

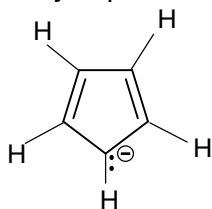
Sol. Octahedral complex with C.N number 6.

47. B

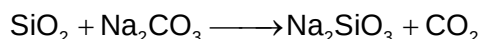
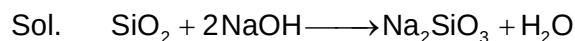
Sol. One Na^+ is removed and remaining three at edge-centres (effective number per unit cell) are present.

48. C

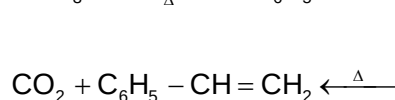
Sol. In cyclopentadienyl anion, each carbon is sp^2 hybridized



49. B

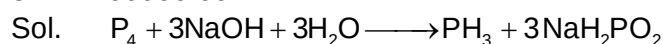


50. D



SECTION – B

51. 00006.00

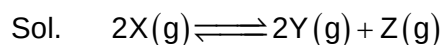


$$X + y = 6$$

52. 00003.00

Sol. The initially formed carbocation will undergo rearrangement. The 2°-carbocation will form an optical isomer when attached with benzene.

53. 00003.00



$$1 - \alpha \quad \alpha \quad \frac{\alpha}{2} \quad \text{Total moles} = 1 + \frac{\alpha}{2} = \frac{2 + \alpha}{2}$$

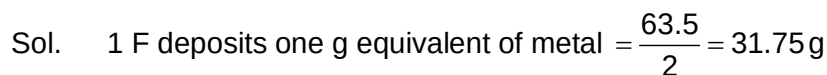
$$K_p = \frac{(P_Y)^2 (P_Z)}{(P_X)^2}$$

$$= \frac{\left(\frac{2\alpha}{2+\alpha}\right)^2 P^2 \cdot \left(\frac{\alpha}{2+\alpha}\right) P}{\left(\frac{2(1-\alpha)}{2+\alpha}\right)^2 P^2}$$

 neglecting α compared to 1 and solving

$$\alpha = \left(\frac{2K_p}{P}\right)^{1/3} \Rightarrow n = 3$$

54. 00031.75



55. 00001.00

Sol. $\text{Slope} = \frac{-\Delta H}{2.303R}$

$$\text{so, } \frac{-\Delta H}{2.303R} = \frac{-1}{4.606}$$

$$\Delta H = 1 \text{ cal}$$

56. 00036.00

Sol. $\text{EAN} = 24(\text{Fe}^{2+})$
 $+12(6 \text{ CN}^-) = 36$

57. -00089.40

Sol. $\text{N}_2 + \frac{1}{2} \text{O}_2 \longrightarrow \text{N} = \text{N} = \text{O}$
 $\text{N} \equiv \text{N} \quad \text{O} = \text{O}$

$$\Delta H = \sum \text{BE}_R - \sum \text{BE}_P$$

$$= \left(946.2 + \frac{497}{2} \right) - (418 + 605.3) = 171.4 \text{ kJ/mol}$$

$$\text{RE} = \Delta H_{\text{exp}} - \Delta H_{\text{cal}} = 82 - 171.4 = -89.4 \text{ kJ}$$

58. 00008.00

Sol. The possible orbitals are 4s and 3p. 4s and 3p contains maximum of two and six electrons.

59. 00009.00

Sol. $x = 6, y = 0, z = 3$

60. 00007.61

(Range 7.61 to 7.64)

	Mass	Moles	Volume
Solute	40 g	40/63	
Solvent	60 g		
Solution	100 g		100/1.2

$$M = \frac{40}{63} \times \frac{1.2}{100} \times 1000$$

$$M = 7.61$$

Mathematics

PART – C

SECTION – A

61. B

$$\begin{aligned} \text{Sol. } f(x) &= \sqrt{(x-1)+4-4\sqrt{x-1}} + \sqrt{(x-1)+9-6\sqrt{x-1}} \\ &= |\sqrt{x-1}-2| + |\sqrt{x-1}-3| \text{ at } x=1.5 \\ f(x) &= (2-\sqrt{x-1}) + (3-\sqrt{x-1}) = 5-2\sqrt{x-1} \\ f'(x) &= \frac{-2}{2\sqrt{x-1}} \\ f'(1.5) &= -\frac{1}{\sqrt{\frac{3}{2}-1}} = -\sqrt{2} \end{aligned}$$

62. C

$$\begin{aligned} \text{Sol. } t_2 &= -t_1 - \frac{2}{t_1} \text{ also } \frac{2at_1}{at_1^2} \times \frac{2at_2}{at_2^2} = -1 \\ \Rightarrow t_1 t_2 &= -4 \\ \therefore -\frac{4}{t_1} &= -t_1 - \frac{2}{t_1} \\ \Rightarrow t_1^2 + 2 &= 4 \text{ and } t_1 = \pm \sqrt{2} \\ \text{so, point can be } &(2a, \pm 2\sqrt{2}a). \end{aligned}$$

63. A

$$\begin{aligned} \text{Sol. } |\vec{a} + \vec{b}| &= 3\lambda^2 + 6 \\ (\lambda + 2)^2 &= 3\lambda^2 + 6 \\ \lambda^2 + 4\lambda + 4 &= 3\lambda^2 + 6 \\ 4\lambda + 4 &= 2\lambda^2 + 6 \\ 4\lambda^2 - 4\lambda + 2 &= 0 \\ \lambda^2 - 2\lambda + 1 &= 0 \\ \lambda &= 1 \end{aligned}$$

64. A

$$\begin{aligned} \text{Sol. } \text{Let the tangent } &\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1 \\ A \equiv (a \cos \theta, 0), B \equiv (0, b \cot \theta) \\ \therefore P \equiv (a \cos \theta, b \cot \theta) &= (h, k) \\ \sec \theta &= \frac{a}{h}, \tan \theta = \frac{b}{k} \\ \therefore \frac{a^2}{h^2} &= \frac{b^2}{k^2} + 1 \end{aligned}$$

$$\therefore \text{Locus is } \frac{a^2}{x^2} - \frac{b^2}{y^2} = 1.$$

65. C

Sol. L_1, L_2 and L_3 are mutually perpendicular lines and all these lines are passing through $P(1, 2, 3)$.

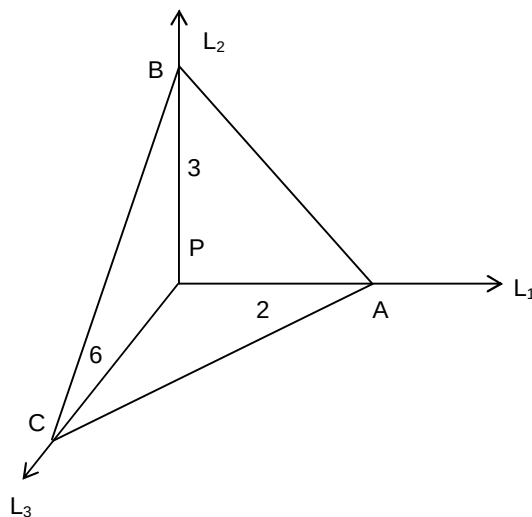
$$V = \frac{1}{6} [\overrightarrow{PA} \cdot \overrightarrow{PB} \times \overrightarrow{PC}]$$

$$= \frac{1}{6} \times 2 \times 3 \times 6 = 6 \text{ cubic units}$$

$$V = \frac{1}{3} (\text{Base area}) (\text{Height})$$

$$6 = \frac{1}{3} \times \sqrt{\left(\frac{1}{2} \cdot 2 \cdot 3\right)^2 + \left(\frac{1}{2} \cdot 3 \cdot 6\right)^2 + \left(\frac{1}{2} \cdot 2 \cdot 6\right)^2} \times d$$

$$\therefore d = \frac{18}{\sqrt{9+81+36}} = \frac{18}{\sqrt{126}} = \frac{6}{\sqrt{14}} \text{ units}$$



66. C

$$\text{Sol. } \frac{1}{\sqrt{x-1}} + \frac{1}{\sqrt{y-1}} + \frac{1}{\sqrt{z-1}} > 0 \Rightarrow x > 1, y >$$

$$1, z > 1$$

x, y, z are in G.P. $\Rightarrow \log x, \log y, \log z$ are in A.P.

$\Rightarrow 2 \log x, \log xy, \log xz$ are in A.P.

$\therefore (\log x^2)^{-1}, (\log xy)^{-1}, (\log xz)^{-1}$ are in H.P.

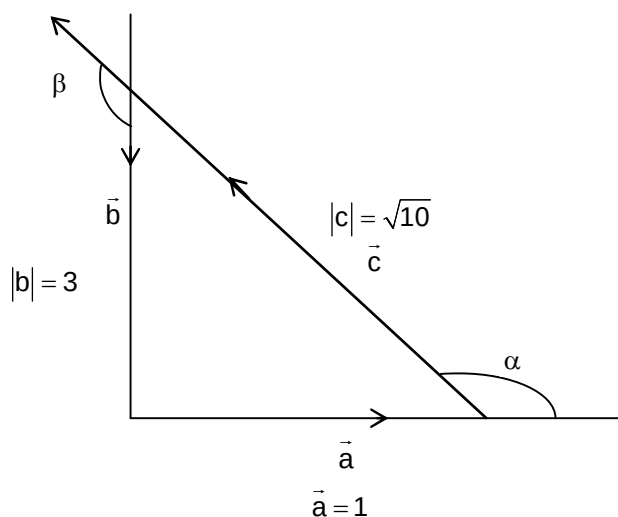
67. C

$$\begin{aligned} \text{Sol. } AB &= A & BA &= B \\ ABA &= A^2 & BAB &= B^2 \\ AB &= A^2 & BA &= B^2 \\ A &= A^2 & B &= B^2 \\ \Rightarrow A^n &= A & B^n &= B \end{aligned}$$

68. C

$$\text{Sol. } \pi - \alpha + \pi - \beta = \frac{\pi}{2}$$

$$\alpha + \beta = \frac{3\pi}{2}$$



69. B

$$\text{Sol. } \frac{\alpha + 2 - 2\beta}{2} = 2 \text{ and } \frac{3\alpha + \beta}{2} = 3$$

$$\therefore \alpha - 2\beta = 2 \text{ and } 3\alpha + \beta = 6$$

$$\therefore 3(2 + 2\beta) + \beta = 6$$

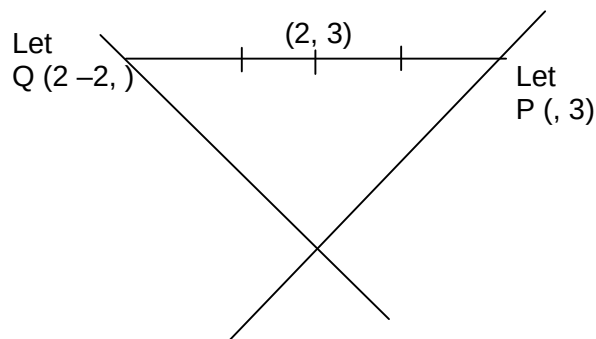
$$7\beta = 0$$

$$\therefore \beta = 0$$

$$\alpha = 0$$

$$\therefore P(2, 6), Q(2, 0)$$

$$\therefore PQ = 6$$



70. A

Sol. Let the coordinates of the mid-point of AB be D (h, k).

Since D lies on $2x - 5y + 18 = 0$,

$$2h - 5k + 18 = 0 \quad \dots(i)$$

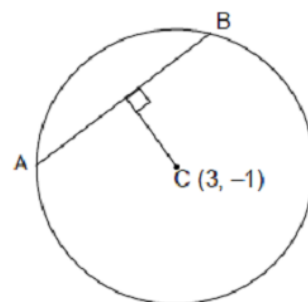
$$\text{Equation of CD is } y + 1 = \frac{5}{2}(x - 3)$$

$$\text{Or } 5x + 2y - 13 = 0 \quad \dots(ii)$$

D lies on CD.

$$\therefore 5h + 2k - 13 = 0 \quad \dots(iii)$$

Solving Equation (i) and (iii), we get $h = 1, k = 4$



71. D

$$\text{Sol. } T_n = \frac{4n + \sqrt{4n^2 - 1}}{\sqrt{2n+1} + \sqrt{2n-1}} \times \frac{\sqrt{2n+1} - \sqrt{2n-1}}{\sqrt{2n+1} - \sqrt{2n-1}}$$

$$T_n = \frac{1}{2} [4n\sqrt{2n+1} - 4n\sqrt{2n-1} + (2n+1)\sqrt{2n-1} - (2n-1)\sqrt{2n+1}]$$

$$T_n = \frac{1}{2} \left[(2n+1)^{3/2} - (2n-1)^{3/2} \right]$$

$$\sum_{n=1}^{40} = \frac{1}{2} \left[3^{3/2} - 1^{3/2} + 5^{3/2} + \dots + 81^{3/2} - 79^{3/2} \right]$$

$$= \frac{1}{2} [729 - 1] = \frac{728}{2} = 364$$

72. D

Sol. ${}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \dots + (-1)^r {}^nC_r = 28$

$$\Rightarrow {}^{n-1}C_0 - ({}^{n-1}C_0 + {}^{n-1}C_1) + ({}^{n-1}C_1 + {}^{n-1}C_2) - ({}^{n-1}C_2 + {}^{n-1}C_3) + \dots$$

$$\dots + (-1)^r ({}^{n-1}C_{r-1} + {}^{n-1}C_r) = 28$$

$$\Rightarrow (-1)^r {}^{n-1}C_r = 28 \Rightarrow r \text{ must be even}$$

$${}^{n-1}C_r = 28 \Rightarrow {}^{n-1}C_r = 7 \times 4 = \frac{7 \times 8}{2} = {}^8C_2 \Rightarrow n-1 = 8 \Rightarrow n = 9$$

73. B

Sol. $x y z y x$, $x < y$; clearly $x \neq 0$

$$\Rightarrow x \in \{1, 2, 3, \dots, 9\} \text{ and } x < y$$

$$\Rightarrow x \neq y \Rightarrow x \in \{1, 2, 3, 4, 5, \dots, 8\} \text{ and } y \in \{2, 3, 4, \dots, 9\}$$

$$\Rightarrow \text{Number of choices for } y = 8 + 7 + 6 + \dots + 1 = 36$$

Also z has 10 choices (any digit)

$$= \text{Total number of numbers} = 36 \times 10 = 360$$

74. A

Sol. Squaring and adding, we get

$$x^2 + y^2 = 1 + 1 + 2\cos(2A - A)$$

$$\therefore \frac{x^2 + y^2 - 2}{2} = \cos A$$

Also $\cos A + 2\cos^2 A - 1 = y$

or $(\cos A + 1)(2\cos A - 1) = y$

Also $\cos A + 2\cos^2 A - 1 = y$

or $(\cos A + 1)(2\cos A - 1) = y$

Put value of $\cos A$ from (1) and get the answer.

75. C

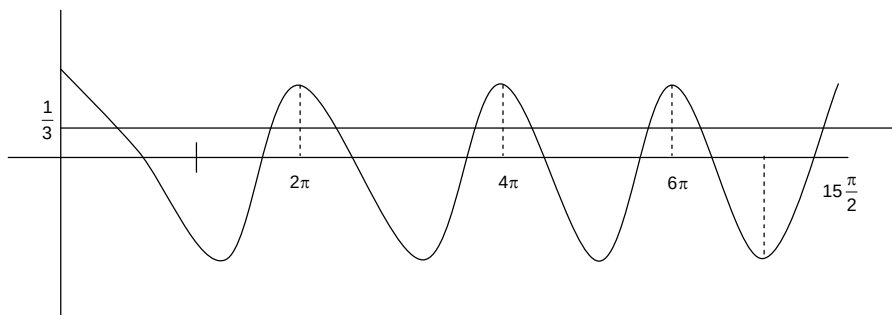
Sol. $3\tan 3x = 3$

$$\tan 3x = 1$$

$$\Rightarrow 3x = n\pi + \frac{\pi}{4}, n \in \mathbb{I}$$

76. D

Sol. $2\sec^2 x - 5\sec x - 3 = 0$



$$(2 \sec x + 1)(\sec x - 3) = 0$$

$$\Rightarrow \sec x = 3$$

$$\cos x = \frac{1}{3}$$

77. D

Sol. Let $y = f(\tan x)$ and $z = g(\sec x)$

$$\frac{dy}{dz} = \frac{f'(\tan x) \cdot \sec^2 x}{g'(\sec x) \cdot \sec x \tan x}$$

$$\left. \frac{dy}{dz} \right|_{x=\frac{\pi}{4}} = \frac{f'(1)\sqrt{2}}{g'(\sqrt{2}) \cdot 1} = \frac{-2\sqrt{2} \cdot \sqrt{2}}{4} = -1$$

78. A

Sol. We have
$$\begin{vmatrix} 1+2x & 1 & 1-x \\ 2-x & 2+x & 3+x \\ x & 1+x & 1-x^2 \end{vmatrix} = 0$$

$$= -2(x - \alpha_1)(x - \alpha_2)(x - \alpha_3)(x - \alpha_4)$$

$$\text{Put } x = 0 \Rightarrow p = \alpha_1 \alpha_2 \alpha_3 \alpha_4 = \frac{1}{2} \Rightarrow 4p = 2$$

79. D

Sol. Planes P_1 and P_2 are perpendicular

$$L: \frac{x+7}{a} = \frac{y-y_1}{-b} = \frac{z-z_1}{c}$$

The above line lies in plane P_2 and perpendicular to plane P_1

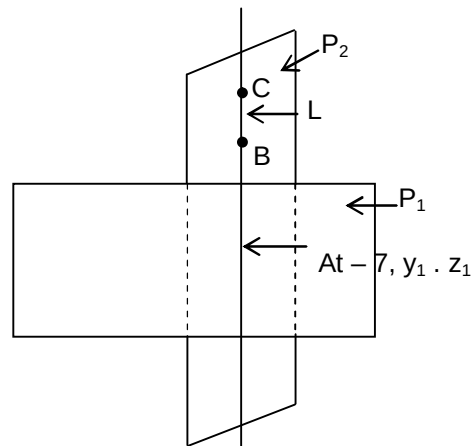
$\therefore A(-7, y_1, z_1)$ lies on both the planes.

$$\therefore -14 - 3y_1 + z_1 = -5 \quad \dots(i)$$

$$-7 - 4y_1 - 14z_1 = 5 \quad \dots(ii)$$

Solving Equation (i) and (ii),

We get $y_1 = -3$ and $z_1 = 0$



$$\therefore L: \frac{x+7}{2} = \frac{y+3}{-3} = \frac{z-0}{1}$$

Now, verify the options.

80. B

Sol. $\vec{a} + \vec{b} + \vec{c} = 0$

$$\vec{a} + \vec{b} = -\vec{c}$$

$$a^2 = b^2 + 2\vec{a} \cdot \vec{b} = c^2$$

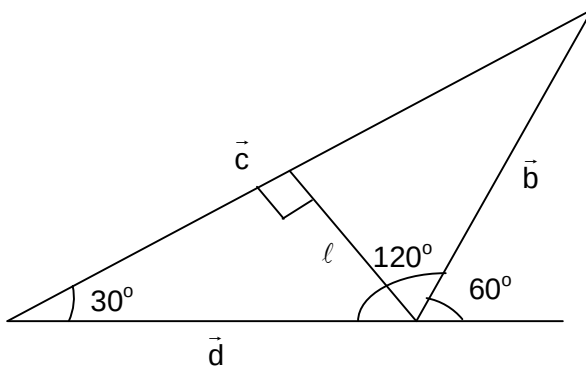
$$a^2 + b^2 - c^2 = -2ab$$

$$\frac{a^2 + b^2 - c^2}{2ab} = \frac{-1}{2}$$

$$\Rightarrow \cos C = \frac{-1}{2} = \angle C = \frac{2\pi}{3}$$

$$|\vec{a}| = |\vec{b}| = 2\sqrt{3}$$

$\therefore$ Area of the triangle $= \frac{1}{2} \times 2\sqrt{3} \cdot 2\sqrt{3} \times \frac{\sqrt{3}}{2} = 3\sqrt{3}$ and length of the median bisecting the vector $\vec{c}$, $\ell = 3 \tan 30^\circ = \sqrt{3}$



SECTION - B

81. 00001.00

Sol. $p \Rightarrow q$ is logically equivalent to $\sim q \Rightarrow \sim p$

$\therefore (p \Rightarrow q) \Leftrightarrow (\sim q \Rightarrow \sim p)$ is a tautology

only (S - 2) is false

82. 00002.00

Sol. Image of Q under in plane

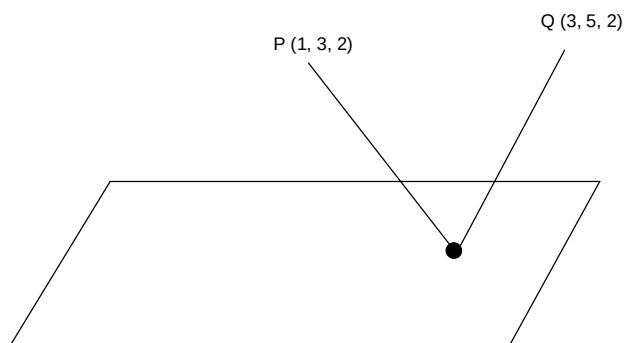
$$\frac{x-3}{1} = \frac{y-5}{2} = \frac{z-2}{1} = -3 \quad (12)$$

$$Q'(-1, -3, -2)$$

Equation 'PQ'

$$\frac{x-1}{2} = \frac{y-3}{6} = \frac{z-2}{4} = \lambda \text{ intersection}$$

with image plane gives R



83. 00005.00

Sol. Let $\sum_{r=0}^{10} {}^{10}C_r \cdot (r-5)^3 = S$

replace r by $10-r$

$$\Rightarrow \sum_{r=10}^0 {}^{10}C_{10-r} \cdot (10-r-5)^3 = S$$

$$\begin{aligned} \Rightarrow \sum_{r=0}^{10} {}^{10}C_r (5-r)^3 &= S \\ \Rightarrow 2S &= 0 \Rightarrow S = 0 \\ \Rightarrow \sum_{r=0}^{10} {}^{10}C_r (r-5)^3 &= 0 \\ \Rightarrow {}^{10}C_0 (-5)^3 + \sum_{r=1}^{10} {}^{10}C_r (r-5)^3 &= 0 \\ \Rightarrow \sum_{r=1}^{10} {}^{10}C_r (r-5)^3 &= 125 \end{aligned}$$

84. 00006.00

Sol. Number of ways in which balls of different colours are drawn
 $= 12 + 3(w) + 4(w) = 12 + 7w$

$${}^3C_1 \times {}^u C_1 \quad {}^3C_1 {}^w C_1 \quad {}^4C_1 \times {}^w C_1$$

$$\frac{12}{12 - 7W} = \frac{2}{9}$$

(Conditional Probability)

$$54 = 12 + 7W$$

$$54 - 12 = 7W$$

$$7W = 42$$

$$W = 6$$

85. 00002.00

Sol. Let $|\vec{a}| = |\vec{b}| = \lambda (> 0)$ and $\vec{a} \cdot \vec{b} = 0$ (Given)

$$\text{Now } |3\vec{a} + 4\vec{b}|^2 = 9\lambda^2 + 16\lambda^2 = 25\lambda^2 \quad \dots\dots\dots(1)$$

$$\text{and } |4\vec{a} - 3\vec{b}|^2 = 16\lambda^2 + 9\lambda^2 = 25\lambda^2 \quad \dots\dots\dots(2)$$

$\therefore$ From equation (1) and (2), we get

$$|3\vec{a} + 4\vec{b}| + |4\vec{a} - 3\vec{b}| = 5\lambda + 5\lambda = 20 \quad \text{(Given)}$$

$$\text{Hence } \lambda = 2 = |\vec{a}| = |\vec{b}|$$

86. 00003.00

Sol. $f(x) = \sin\left(\left[x\right]\frac{\pi}{2}\right) \text{sgn}(x-1)(x-2)$

$$= \begin{cases} \sin\left(\left[x\right]\frac{\pi}{2}\right) & x < 1 \text{ or } x > 2 \\ -\sin\left(\left|x\right|\frac{\pi}{2}\right) & 1 < x < 2 \\ 0 & x = 1, 2 \end{cases}$$

$$\begin{cases} 0 & x \in [0,1) \\ 0 & x = 1 \\ -1 & x \in (1,2) \\ 0 & x = 2 \\ 0 & x \in (2,3) \\ -1 & x = 3 \end{cases} = \begin{cases} 0 & 0 \leq x \leq 1 \\ -1 & 1 < x < 2 \\ 0 & 2 \leq x < 3 \\ 1 & x = 3 \end{cases}$$

$\Rightarrow$ function is discontinuous at $x = 1, 2, 3$

87. 00190.00

Sol. $B^2 = 1$

$$AB = \begin{bmatrix} a & x & p \\ y & q & b \\ r & c & z \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} p & x & a \\ b & q & y \\ z & c & r \end{bmatrix}$$

$$AB = AB^3 = \dots\dots\dots = AB^{19} = \begin{bmatrix} p & x & a \\ b & q & y \\ z & c & r \end{bmatrix}$$

$$\text{tr.}(AB + AB^3 + \dots\dots\dots + AB^{19}) = 210$$

$$\Rightarrow 10(p + q + r) = 210 \Rightarrow p + q + r = 21, p, q, r \in \mathbb{N}$$

$$p' + q' + r' = 18, p', q', r' \in \mathbb{W}$$

$$\therefore \text{Number of ordered triplets } (p, q, r) = {}^{20}C_r = \frac{20 \times 19}{2} = 190$$

88. 00011.00

Sol. $(x^2 + 12x + 35)(x^2 + 10x + 24) = 504$

$$(x - 5)(x - 7)(x + 6)(x + 4) = 504$$

$$(x - 7)(x + 6)(x - 5)(x + 4) = 504$$

$$(x^2 - x - 42)(x^2 - x - 20) = 504$$

$$x^2 - x = t$$

$$(t - 42)(t - 20) = 504$$

$$x^2 - x = 6$$

$$t^2 - 62t + 840 - 504 = 0$$

$$x^2 - x - 6 = 0$$

$$t^2 - 62t + 336 = 0$$

$$x = 3 \text{ and } -2$$

$$(t - 6)(t - 56) =$$

$$x^2 - x = 56$$

$$t = 6 \text{ or } 56 \quad x^2 - x - 56 = 0$$

$$(x - 8)(x + 7) = 0$$

$$x = -78$$

$$x_1 + x_2 = 8 + 3 = 11$$

89. 00021.00

Sol. $B = \text{adj}(A) \Rightarrow B = |A| A^{-1}$

$$B = 3A^{-1}, B^{-1} = \frac{1}{3}A$$

$$f(x) = \frac{1}{3}(x^3 - 6x^2 + 9x + 9)$$

$$f'(x) = x^2 - 4x + 3$$

$$\text{Global maximum at } x = 6 \Rightarrow \frac{1}{3}(72 - 72 + 54 + 9) = 21$$

90. 00144.00

Sol. $\det.(A^2 + A) = \det\left[\left(P^{-1}DP(P^{-1}DP) + P^{-1}DP\right)\right]$

Or $\det.(P^{-1}D^2P + P^{-1}DP)$

Or $\det(P^{-1}(D^2 + D)P)$

Or $\det.(P^{-1}P) \cdot \det(D^2 + D)$

Or $\det.(I) \cdot \det(D) \cdot \det.(D + I)$

Or $1^2 \cdot (1 \cdot 2 \cdot 3) \cdot (2 \cdot 3 \cdot 4) = 6 \cdot 24 = 144$